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Pressure gauge and fluid pressure system having a fitting structure arrangement

Abstract

A pressure gauge of a fluid pressure system includes a measurement unit, a case, a display unit, and an adapter. The adapter includes an attachment hole. The case includes an attachment convex portion which can be fitted into the attachment hole. The display unit includes a display surface oriented in a direction intersecting an axis of the attachment convex portion. The attachment convex portion and the attachment hole have a fitting structure capable of changing the relative position between the attachment convex portion and the adapter in the circumferential direction about the axis of the attachment convex portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2021-204698 filed on Dec. 17, 2021, the contents of which are incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

(2) The present invention relates to a pressure gauge and a fluid pressure system.

Description of the Related Art

(3) For example, JP H10-206262 A discloses a pressure gauge attached to a fluid pressure device.

The pressure gauge includes a case that accommodates a measurement unit for measuring the pressure of a fluid, and a display unit provided on the case. The case is fixed to the fluid pressure device by a screw member. The display unit has a display surface oriented in the axial direction of the screw member.

SUMMARY OF THE INVENTION

(4) In the above-described pressure gauge, the orientation of the display surface cannot be changed when the pressure gauge is attached to the fluid pressure device. Therefore, depending on the user's usage, the display surface of the pressure gauge may be oriented in a direction in which the display surface is difficult to view.

(5) The present invention has the object of solving the aforementioned problem.

(6) According to an aspect of the present invention, there is provided a pressure gauge comprising: a case configured to accommodate a measurement unit configured to measure a pressure of a fluid; a display unit provided in the case and configured to display the pressure measured by the measurement unit; and an adapter attached to a fluid pressure device, wherein the adapter includes an attachment hole, the case includes an attachment convex portion configured to be fitted into the attachment hole, the display unit includes a display surface oriented in a direction intersecting an axis of the attachment convex portion, and the attachment convex portion and the attachment hole have a fitting structure that allows a relative position between the attachment convex portion and the adapter in a circumferential direction about the axis of the attachment convex portion to be changed.

(7) According to another aspect of the present invention, there is provided a fluid pressure system comprising a fluid pressure device, and a pressure gauge attached to the fluid pressure device, wherein the pressure gauge is the above-described pressure gauge.

(8) According to the present invention, the display surface is oriented in the direction intersecting the axis of the attachment convex portion. Further, the attachment convex portion and the attachment hole have a fitting structure capable of changing the relative position between the attachment convex portion and the adapter in the circumferential direction about the axis of the attachment convex portion. Therefore, the orientation of the display surface can be easily changed. As a result, it is possible to orient the display surface in a direction in which the display surface is easily visible, according to the user's usage.

(9) The above and other objects, features, and advantages of the present invention will become more apparent from the following description when taken in conjunction with the accompanying drawings, in which a preferred embodiment of the present invention is shown by way of illustrative example.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a fluid pressure system according to an embodiment of the present invention;

(2) FIG. 2 is an exploded perspective view of a pressure gauge of FIG. 1;

(3) FIG. 3 is a longitudinal cross-sectional view of the pressure gauge of FIG. 2;

(4) FIG. 4 is a longitudinal cross-sectional view taken along line IV-IV of FIG. 3;

(5) FIG. 5 is a front view of a pressure gauge main body of FIG. 2;

(6) FIG. 6 is a bottom view of the pressure gauge main body of FIG. 2;

(7) FIG. 7 is a transverse cross-sectional view of an adapter of FIG. 2;

(8) FIG. 8 is a transverse cross-sectional view taken along line VIII-VIII of FIG. 3;

(9) FIG. 9A is a perspective view of the fluid pressure system according to a first example in which the orientation of a display surface of the pressure gauge of FIG. 1 is changed;

- (10) FIG. 9B is a perspective view of the fluid pressure system according to a second example in which the orientation of the display surface of the pressure gauge of FIG. 1 is changed;
- (11) FIG. 10 is an exploded perspective view of a pressure gauge according to a modification;
- (12) FIG. 11 is a front view of a pressure gauge main body of FIG. 10;
- (13) FIG. 12 is a bottom view of the pressure gauge main body of FIG. 10; and
- (14) FIG. 13 is a transverse cross-sectional view of the pressure gauge of FIG. 10.

DESCRIPTION OF THE INVENTION

- (15) As shown in FIG. 1, a fluid pressure system **10** according to an embodiment of the present invention includes a fluid pressure device **12**, and a pressure gauge **14** attached to the fluid pressure device **12**. The fluid pressure device **12** is, for example, a regulator (pressure regulator) for regulating the pressure of a fluid. The regulator includes a valve main body **16** and a handle **18**.
- (16) The valve main body **16** is formed in a rectangular parallelepiped shape. The handle **18** that is operated by being rotated is attached to the valve main body **16**. The valve main body **16** includes a plurality of ports **20** through which the fluid flows. The plurality of ports **20** include input ports and output ports. The pressure gauge **14** is attached to an outer surface of the valve main body **16** that is oriented in a direction orthogonal to the axial direction of the handle **18**. The handle **18** adjusts the set pressure of the fluid delivered from the fluid pressure device **12**. The fluid pressure device **12** to which the pressure gauge **14** is attached is not limited to a regulator.
- (17) As shown in FIG. 2, the pressure gauge **14** includes a pressure gauge main body **22**, an adapter **24**, and a retaining member **26**. The pressure gauge main body **22** includes a measurement unit **28**, a display unit **30**, and a case **32**. The measurement unit **28** measures the pressure of the fluid. The display unit **30** displays the pressure measured by the measurement unit **28**.
- (18) In FIGS. 2 and 3, the display unit **30** includes a display main body **34** and a cover **36**. The display main body **34** is a square plate-shaped portion. A display surface **37** is provided on one surface of the display main body **34**. The display surface **37** is provided with a scale (not shown). A pointer **38** is rotatably attached to a central portion of the display surface **37**. The measurement unit **28** is attached to another surface of the display main body **34**.
- (19) As shown in FIG. 2, the display main body **34** includes two insertion holes **40**. The two insertion holes **40** are respectively located at diagonal corners of the display main body **34**. A bolt **41** for attaching the display main body **34** to the case **32** is inserted through each insertion hole **40**.
- (20) The cover **36** is attached to one surface of the display main body **34**. The cover **36** includes a frame portion **42** that covers an outer peripheral portion of the display surface **37** of the display main body **34**, and a transparent portion **44** that covers the display surface **37**.
- (21) As shown in FIGS. 2 to 4, the case **32** is an integrally molded product made of a resin material. The case **32** includes a hollow case main body **46**, a fluid introduction portion **48**, and a plurality of reinforcing ribs **52**. The case main body **46** includes a peripheral wall portion **54** having a square tubular shape, and an end wall portion **56** (see FIG. 3). The peripheral wall portion **54** extends in one direction.
- (22) In the following description, the extending direction of the peripheral wall portion **54** may be referred to as an arrow X direction, a width direction of the peripheral wall portion **54** may be referred to as an arrow Y direction, and a height direction of the peripheral wall portion **54** may be referred to as an arrow Z direction. The arrow Z direction lies along the thickness direction of the plate-shaped adapter **24**.
- (23) The outer surface of the peripheral wall portion **54** has a substantially square transverse cross-sectional shape (see FIG. 4). In FIGS. 2 and 3, one end portion (an end portion on an arrow X1 direction side) of the peripheral wall portion **54** includes an attachment surface **58** to which the display unit **30** is attached. The attachment surface **58** is a flat surface oriented in the arrow X1 direction. The outer shape of the attachment surface **58** is square. In other words, the outer shape of the attachment surface **58** is the same as the outer shape of the display main body **34**. Further, the outer size of the attachment surface **58** is the same as the outer size of the display main body **34**. A

hole **60** for inserting the measurement unit **28** into an internal space **S** of the peripheral wall portion **54** is formed in the attachment surface **58**.

(24) In FIG. **4**, each of the four corners of the inner surface of the peripheral wall portion **54** includes a bulging portion **62** bulging inward of the peripheral wall portion **54**. A nut **64**, into which the bolt **41** for fixing the display unit **30** to the case main body **46** is screwed, is insert-molded in each of the bulging portions **62**.

(25) As shown in FIG. **2**, the display unit **30** is attached to the case **32** by screwing the two bolts **41**, inserted through the two insertion holes **40** of the display main body **34**, into the two nuts **64**. The display unit **30** can be attached to the case **32** such that the relative position between the display unit **30** and the peripheral wall portion **54** in the circumferential direction about the rotation axis of the pointer **38** can be changed. In other words, the display unit **30** can be attached to the case **32** such that the orientation of an upper end surface **35** of the display main body **34** can be changed.

(26) Specifically, the display unit **30** can be attached to the case **32** such that the upper end surface **35** of the display main body **34** is oriented in an arrow **Z2** direction. Further, the display unit **30** can be attached to the case **32** such that the upper end surface **35** of the display main body **34** is oriented in an arrow **Y1** direction. Furthermore, the display unit **30** can be attached to the case **32** such that the upper end surface **35** of the display main body **34** is oriented in an arrow **Z1** direction. Moreover, the display unit **30** can be attached to the case **32** such that the upper end surface **35** of the display main body **34** is oriented in an arrow **Y2** direction. Accordingly, when the display unit **30** is attached to the case **32**, the orientation of the upper end surface **35** of the display main body **34** can be changed according to the user's usage.

(27) In FIG. **3**, the end wall portion **56** is located at another end portion (an end portion on an arrow **X2** side) of the peripheral wall portion **54**. The end wall portion **56** covers the internal space **S** of the peripheral wall portion **54** from the arrow **X2** direction.

(28) As shown in FIGS. **3** and **4**, an introduction flow path **66** for guiding the fluid of the fluid pressure device **12** to the measurement unit **28** is formed in the fluid introduction portion **48**. The fluid introduction portion **48** includes an attachment convex portion **69**, a first introduction pipe portion **70**, a second introduction pipe portion **72**, and a connection pipe portion **74**. The attachment convex portion **69** protrudes outward of the peripheral wall portion **54** (in the arrow **Z1** direction) from a support wall portion **68** of the peripheral wall portion **54**, the support wall portion **68** being located in a direction (on the arrow **Z1** direction side) orthogonal to the rotation axis of the pointer **38**.

(29) The first introduction pipe portion **70** extends in the arrow **Z1** direction from a protruding end surface of the attachment convex portion **69**. An extended end portion of the first introduction pipe portion **70** is inserted into the valve main body **16** in a state where the pressure gauge **14** is attached to the fluid pressure device **12**.

(30) The second introduction pipe portion **72** extends from the support wall portion **68** in a direction (the arrow **Z2** direction) opposite to the extending direction of the first introduction pipe portion **70**. The second introduction pipe portion **72** extends to the internal space **S** of the peripheral wall portion **54**. The inner hole of the second introduction pipe portion **72** communicates with the inner hole of the first introduction pipe portion **70**. The second introduction pipe portion **72** is connected to the inner surface of the end wall portion **56** (see FIG. **3**).

(31) The connection pipe portion **74** protrudes in the arrow **X1** direction from the inner surface of the end wall portion **56** toward the measurement unit **28**. An extended end portion of the second introduction pipe portion **72** is connected to the connection pipe portion **74**. The inner hole of the connection pipe portion **74** communicates with the inner hole of the second introduction pipe portion **72**. The measurement unit **28** is airtightly connected to a protruding end of the connection pipe portion **74**.

(32) As shown in FIG. **3**, the display surface **37** of the display unit **30** is oriented in a direction intersecting an axis **Ax** of the attachment convex portion **69**. In other words, the display surface **37**

of the display unit **30** is oriented in a direction orthogonal to the axis Ax of the attachment convex portion **69**. The attachment convex portion **69** is provided with the introduction flow path **66** (see FIGS. 3 and 4).

(33) As shown in FIG. 6, the attachment convex portion **69** has a polygonal outer attachment surface **76**. In other words, the attachment convex portion **69** has the outer attachment surface **76** having an octagonal shape. Specifically, the outer attachment surface **76** is formed in a regular octagonal shape when viewed from the arrow Z1 direction. That is, the outer attachment surface **76** has eight flat surfaces **77**. The attachment convex portion **69** is 45° rotationally symmetric about the axis Ax of the attachment convex portion **69**. An annular groove **78** extending in the circumferential direction about the axis Ax of the attachment convex portion **69** is formed in an intermediate portion of the outer attachment surface **76** in the arrow Z direction (see FIGS. 3 to 5).

(34) The plurality of reinforcing ribs **52** extend from the protruding end surface of the attachment convex portion **69** toward the extended end portion of the first introduction pipe portion **70**. In the present embodiment, the case **32** includes eight reinforcing ribs **52**. The reinforcing ribs **52** are respectively connected to a plurality of corner portions of the outer attachment surface **76**. The plurality of reinforcing ribs **52** are arranged at equal intervals in the circumferential direction about the axis Ax of the attachment convex portion **69**. The shape, size, number, and the like of the reinforcing ribs **52** can be set as appropriate.

(35) As shown in FIG. 2, the adapter **24** includes a quadrangular plate-shaped adapter main body **79**, and a protruding portion **81**. Specifically, the adapter main body **79** is formed in a square shape. The adapter main body **79** has a first side surface **80a**, a second side surface **80b**, a third side surface **80c**, and a fourth side surface **80d**. The first side surface **80a** and the second side surface **80b** extend in the same direction. The third side surface **80c** and the fourth side surface **80d** extend in a direction orthogonal to the extending direction of the first side surface **80a** and the second side surface **80b**.

(36) The adapter main body **79** includes four insertion holes **84**. The four insertion holes **84** are located at four corners of the adapter main body **79**, respectively. The adapter main body **79** is 90° rotationally symmetric about the center line of the adapter main body **79** when viewed from the thickness direction. Bolts **86** for attaching the adapter **24** to the fluid pressure device **12** are inserted through two insertion holes **84** that are diagonally positioned among the four insertion holes **84**.

(37) The protruding portion **81** protrudes in the thickness direction from one surface of the adapter main body **79**. The protruding portion **81** extends in an annular shape.

(38) In FIGS. 7 and 8, an attachment hole **88** into which the attachment convex portion **69** can be fitted is formed in a central portion of the adapter main body **79**. The adapter main body **79** includes an inner attachment surface **90** surrounding the attachment hole **88**. The inner attachment surface **90** is formed in a polygonal shape. The inner attachment surface **90** is close to or in contact with the outer attachment surface **76** such that rotation of the attachment convex portion **69** about the axis Ax is restricted in a state in which the attachment convex portion **69** is fitted into the attachment hole **88** (see FIG. 8). The inner attachment surface **90** includes two first flat surfaces **92**, two second flat surfaces **94**, and four connecting surfaces **96**.

(39) Each first flat surface **92** extends along the extending direction of the first side surface **80a** and the second side surface **80b** of the adapter **24**. The two first flat surfaces **92** are positioned to face each other. The length of each first flat surface **92** in the extending direction thereof is longer than the width of each flat surface **77** of the outer attachment surface **76**. The two first flat surfaces **92** have the same length in the extending direction thereof.

(40) Each second flat surface **94** extends along the extending direction of the third side surface **80c** and the fourth side surface **80d** of the adapter **24**. In other words, each second flat surface **94** extends in a direction orthogonal to the extending direction of each first flat surface **92**. The two second flat surfaces **94** are positioned to face each other. The length of each second flat surface **94** along the extending direction thereof is shorter than the length of each first flat surface **92** along the

extending direction thereof. The two second flat surfaces **94** have the same length in the extending direction thereof.

(41) The four connecting surfaces **96** each connect an end portion of the first flat surface **92** and an end portion of the second flat surface **94** to each other. Each connecting surface **96** includes a first connecting flat surface **98** and a second connecting flat surface **100**. The first connecting flat surface **98** extends from the end portion of the first flat surface **92** toward the attachment hole **88**. The first connecting flat surface **98** extends along the extending direction of the second flat surface **94**.

(42) The second connecting flat surface **100** extends from an extended end portion of the first connecting flat surface **98** to the end portion of the second flat surface **94**. The second connecting flat surface **100** extends in a direction intersecting the extending direction of the first flat surface **92** and the extending direction of the second flat surface **94**. The length of the second connecting flat surface **100** along the extending direction thereof is shorter than the length of the second flat surface **94** along the extending direction thereof. The length of the second connecting flat surface **100** along the extending direction thereof is shorter than the width of each flat surface **77** of the outer attachment surface **76** (see FIG. 8).

(43) In FIG. 8, in a fitted state in which the attachment convex portion **69** is fitted into the attachment hole **88**, the eight flat surfaces **77** of the attachment convex portion **69** face the two first flat surfaces **92**, the two second flat surfaces **94**, and the four second connecting flat surfaces **100**. In other words, in the fitted state, the four first connecting flat surfaces **98** are spaced apart from the outer attachment surface **76**.

(44) In the following description, among the four second connecting flat surfaces **100**, two second connecting flat surfaces **100** extending from the second flat surface **94** toward the first side surface **80a** may be referred to as “second connecting flat surfaces **100a**”. Further, among the four second connecting flat surfaces **100**, two second connecting flat surfaces **100** extending from the second flat surface **94** toward the second side surface **80b** may be referred to as “second connecting flat surfaces **100b**”.

(45) An insertion hole **102** into which the retaining member **26** is inserted is formed in the adapter **24**. The insertion hole **102** includes two first holes **104**, two second holes **106**, and a communication hole **108**. The two first holes **104** extend linearly from the first side surface **80a** to the two second connecting flat surfaces **100a** along the extending direction of the third side surface **80c** and the fourth side surface **80d**. The two first holes **104** extend in parallel to each other. Each first hole **104** opens to the first side surface **80a**. Each first hole **104** includes a first opening portion **110** that opens to the second connecting flat surface **100a**.

(46) The two second holes **106** extend linearly from the second side surface **80b** to the two second connecting flat surfaces **100b** along the extending direction of the third side surface **80c** and the fourth side surface **80d**. The two second holes **106** extend in parallel to each other. The transverse cross-sectional shape of each second hole **106** is the same as the transverse cross-sectional shape of each first hole **104**. Each second hole **106** opens to the second side surface **80b**. Each second hole **106** includes a second opening portion **112** that opens to the second connecting flat surface **100b**. Each second connecting flat surface **100b** is inclined with respect to the extending direction of the second hole **106**. Therefore, the area of the second opening portion **112** is larger than the cross-sectional area of the intermediate portion of the second hole **106**.

(47) One of the second holes **106** extends on an extension line of one of the first holes **104**. The other of the second holes **106** extends on an extension line of the other of the first holes **104**. The first opening portion **110** faces the second opening portion **112**. The communication hole **108** connects end portions of the two first holes **104** to each other. The communication hole **108** opens to the first side surface **80a**.

(48) In FIGS. 2 and 8, the retaining member **26** extends in a line shape. Specifically, the retaining member **26** is a clip formed in a U shape. The retaining member **26** is formed by bending a metal

rod into a U shape. The retaining member **26** is inserted into the insertion hole **102**, thereby preventing the attachment convex portion **69** from coming out of the attachment hole **88**.

(49) The retaining member **26** includes two locking portions **114** and a connecting portion **116**. The two locking portions **114** extend in parallel to each other. The connecting portion **116** connects end portions of the two locking portions **114** to each other. Each locking portion **114** is positioned so as to straddle the first hole **104**, the annular groove **78**, and the second hole **106**. The connecting portion **116** is located in the communication hole **108**.

(50) Next, a procedure for attaching the pressure gauge **14** to the fluid pressure device **12** will be described. First, the user attaches the adapter **24** to the fluid pressure device **12** with the two bolts **86**. In this case, if the adapter **24** is attached to the fluid pressure device **12** such that the first side surface **80a** is oriented in the direction of gravity, there is a possibility that the retaining member **26** falls. In the present embodiment, since the four insertion holes **84** are provided at the respective corner portions of the adapter main body **79**, the user can attach the adapter **24** to the fluid pressure device **12** such that the first side surface **80a** is not oriented in the direction of gravity.

(51) Thereafter, the attachment convex portion **69** is fitted into the attachment hole **88**. Specifically, the attachment convex portion **69** and the attachment hole **88** have a fitting structure capable of changing the relative position between the attachment convex portion **69** and the adapter **24** in the circumferential direction about the axis Ax of the attachment convex portion **69**. Specifically, the orientation of the display surface **37** in a state in which the attachment convex portion **69** is fitted into the attachment hole **88** can be changed at intervals of 45° in the circumferential direction about the axis Ax of the attachment convex portion **69**.

(52) That is, for example, as shown in FIG. **1**, the attachment convex portion **69** can be fitted into the attachment hole **88** in a state where the display surface **37** is oriented in a direction in which the handle **18** protrudes from the valve main body **16**. Further, for example, as shown in FIG. **9A**, the attachment convex portion **69** can be fitted into the attachment hole **88** in a state where the display surface **37** is oriented in a direction inclined by 45° in the circumferential direction about the axis Ax of the attachment convex portion **69** with respect to the direction in which the handle **18** protrudes from the valve main body **16**. Furthermore, for example, as shown in FIG. **9B**, the attachment convex portion **69** can be fitted into the attachment hole **88** in a state where the display surface **37** is oriented in a direction inclined by 90° in the circumferential direction about the axis Ax of the attachment convex portion **69** with respect to the direction in which the handle **18** protrudes from the valve main body **16**. That is, it is possible to easily change the orientation of the display surface **37** according to the user's usage.

(53) After the attachment convex portion **69** is fitted into the attachment hole **88**, the retaining member **26** is inserted into the insertion hole **102** of the adapter **24**. Specifically, as shown in FIG. **8**, the two locking portions **114** of the retaining member **26** are inserted into the annular groove **78** from the two first holes **104**. Then, the distal end portion of each locking portion **114** is inserted into the second opening portion **112**. In this case, since the area of the second opening portion **112** is larger than the cross-sectional area of the second hole **106**, each locking portion **114** is smoothly inserted into the second opening portion **112**. When the insertion of the retaining member **26** into the insertion hole **102** is completed, the attachment of the pressure gauge **14** to the fluid pressure device **12** is completed.

(54) The pressure gauge **14** according to the present embodiment has the following effects.

(55) According to the present embodiment, the display surface **37** is oriented in a direction intersecting the axis Ax of the attachment convex portion **69**. Further, the attachment convex portion **69** and the attachment hole **88** have a fitting structure capable of changing the relative position between the attachment convex portion **69** and the adapter **24** in the circumferential direction about the axis Ax of the attachment convex portion **69**. Therefore, the orientation of the display surface **37** can be easily changed. As a result, the display surface **37** can be oriented in a direction in which the display surface **37** is easily visible, according to the user's usage.

(56) The adapter **24** includes the inner attachment surface **90** surrounding the attachment hole **88**. The attachment convex portion **69** includes the polygonal outer attachment surface **76**. The inner attachment surface **90** restricts rotation of the attachment convex portion **69** about the axis Ax relative to the adapter **24** in the fitted state in which the attachment convex portion **69** is fitted into the attachment hole **88**.

(57) According to such a configuration, it is possible to prevent the orientation of the display surface **37** from being changed when the user unintentionally touches the pressure gauge **14**.

(58) The inner attachment surface **90** is formed in a polygonal shape.

(59) According to such a configuration, rotation of the attachment convex portion **69** relative to the adapter **24** in the fitted state can be restricted with a simple configuration.

(60) The attachment convex portion **69** is 45° rotationally symmetric about the axis Ax of the attachment convex portion **69**.

(61) According to such a configuration, the orientation of the display surface **37** can be changed at intervals of 45° in the circumferential direction about the axis Ax of the attachment convex portion **69**.

(62) The outer attachment surface **76** is formed in an octagonal shape. The inner attachment surface **90** includes a plurality of flat surfaces facing the plurality of flat surfaces **77** of the outer attachment surface **76** in the fitted state.

(63) According to such a configuration, the orientation of the display surface **37** can be changed at intervals of 45° in the circumferential direction about the axis Ax of the attachment convex portion **69** with a simple configuration.

(64) The pressure gauge **14** includes the retaining member **26** having a line shape and used for preventing the attachment convex portion **69** from coming out of the attachment hole **88**. The outer attachment surface **76** includes the annular groove **78** extending in the circumferential direction about the axis Ax of the attachment convex portion **69**. The adapter **24** includes the insertion hole **102** communicating with the annular groove **78** in the fitted state. The retaining member **26** is inserted into the insertion hole **102** and the annular groove **78** so as to straddle the insertion hole **102** and the annular groove **78**.

(65) According to such a configuration, it is possible to prevent the attachment convex portion **69** from coming out of the attachment hole **88** with a simple configuration.

(66) The adapter **24** includes the first holes **104** and the second holes **106**. Each of the first holes **104** extends linearly from the first side surface **80a** of the adapter **24** to the inner attachment surface **90**. Each of the second holes **106** is located on an extension line of the first hole **104** and extends linearly from the inner attachment surface **90**. The second hole **106** opens to the second connecting flat surface **100b** of the inner attachment surface **90**, the second connecting flat surface **100b** extending in a direction inclined with respect to the extending direction of the second hole **106**. The retaining member **26** is inserted into the first holes **104**, the annular groove **78**, and the second holes **106** so as to straddle the first holes **104**, the annular groove **78**, and the second holes **106**.

(67) According to such a configuration, the area of the second opening portion **112** that is a part of the second hole **106** and opens to the second connecting flat surface **100b** can be made larger than the cross-sectional area of the second hole **106**. As a result, after the retaining member **26** is passed through the annular groove **78** from the first holes **104**, the distal end portions of the retaining member **26** can be smoothly inserted into the second opening portions **112**. In addition, since the area of the second opening portion **112** can be increased without chamfering the edge portion of the second opening portion **112**, the manufacturing cost of the adapter **24** can be suppressed.

(68) The fluid pressure system **10** may include a pressure gauge **120** according to a modification shown in FIG. **10** instead of the pressure gauge **14** described above. In the modification, the same reference numerals as those of the above-described embodiment denote the same components. Further, in the modification, description of the same configuration as that of the above-described

embodiment will be omitted.

(69) As shown in FIG. 10, the pressure gauge 120 includes a pressure gauge main body 122, an adapter 124, and the retaining member 26. The pressure gauge main body 122 includes the measurement unit 28, the display unit 30, and a case 126. As shown in FIGS. 10 to 12, a fluid introduction portion 127 of the case 126 includes an attachment convex portion 128. The attachment convex portion 128 includes an outer attachment surface 130 having a circular shape (perfect circular shape). The annular groove 78 is formed in the outer attachment surface 130.

(70) As shown in FIGS. 10 and 13, an adapter main body 125 of the adapter 124 is provided with an attachment hole 132 into which the attachment convex portion 128 can be fitted. The adapter 124 includes an inner attachment surface 134 surrounding the attachment hole 132. The inner attachment surface 134 is formed in a circular shape (perfect circular shape). The inner attachment surface 134 faces the outer attachment surface 130 in a fitted state in which the attachment convex portion 128 is fitted into the attachment hole 132. In the fitted state, the inner attachment surface 134 is close to or in contact with the outer attachment surface 130 such that the attachment convex portion 128 is rotatable about the axis Ax.

(71) In the present modification, the adapter 124 includes the circular inner attachment surface 134 surrounding the attachment hole 132. The attachment convex portion 128 includes the circular outer attachment surface 130. The outer attachment surface 130 and the inner attachment surface 134 face each other in the fitted state in which the attachment convex portion 128 is fitted into the attachment hole 132.

(72) According to such a configuration, the orientation of the display surface 37 can be continuously changed in the circumferential direction about the axis Ax of the attachment convex portion 128. Therefore, the display surface 37 can be oriented in a direction in which the display surface 37 is more easily visible, according to the user's usage.

(73) The attachment convex portion 128 is rotatable about the axis Ax of the attachment convex portion 128 relative to the adapter 124 in the fitted state.

(74) According to such a configuration, the orientation of the display surface 37 can be adjusted in the fitted state in which the attachment convex portion 128 is fitted into the attachment hole 132.

(75) The present invention is not limited to the embodiment described above, and various configurations may be adopted therein without deviating from the essence and gist of the present invention.

(76) The present embodiment discloses the following contents.

(77) The above embodiment discloses the pressure gauge (14, 120) including: the case (32, 126) configured to accommodate the measurement unit (28) configured to measure the pressure of the fluid; the display unit (30) provided in the case and configured to display the pressure measured by the measurement unit; and the adapter (24, 124) attached to the fluid pressure device (12), wherein the adapter includes the attachment hole (88, 132), the case includes the attachment convex portion (69, 128) configured to be fitted into the attachment hole, the display unit includes the display surface (37) oriented in the direction intersecting the axis (Ax) of the attachment convex portion, and the attachment convex portion and the attachment hole have the fitting structure that allows the relative position between the attachment convex portion and the adapter in the circumferential direction about the axis of the attachment convex portion to be changed.

(78) In the above-described pressure gauge, the adapter may include the inner attachment surface (90) configured to surround the attachment hole, the attachment convex portion may include the outer attachment surface (76) having a polygonal shape, and the inner attachment surface may restrict rotation of the attachment convex portion about the axis relative to the adapter in the fitted state in which the attachment convex portion is fitted into the attachment hole.

(79) In the above-described pressure gauge, the inner attachment surface may be formed in a polygonal shape.

(80) In the above-described pressure gauge, the attachment convex portion may be 45° rotationally

symmetric about the axis of the attachment convex portion.

(81) In the above-described pressure gauge, the outer attachment surface may be formed in an octagonal shape, and inner attachment surface may include the plurality of flat surfaces facing the plurality of flat surfaces (77) of the outer attachment surface in the fitted state.

(82) In the above-described pressure gauge, the adapter may include the inner attachment surface (134) having a circular shape and configured to surround the attachment hole, the attachment convex portion may include the outer attachment surface (130) having a circular shape, and the inner attachment surface may face the outer attachment surface in the fitted state in which the attachment convex portion is fitted into the attachment hole.

(83) In the above-described pressure gauge, the attachment convex portion may be rotatable about the axis of the attachment convex portion relative to the adapter in the fitted state.

(84) In the above-described pressure gauge, the pressure gauge may include the retaining member (26) having a line shape and configured to prevent the attachment convex portion from coming out of the attachment hole, the outer attachment surface may include the annular groove (78) extending in the circumferential direction about the axis of the attachment convex portion, the adapter may include the insertion hole (102) communicating with the annular groove in the fitted state, and the retaining member may be inserted into the insertion hole and the annular groove so as to straddle the insertion hole and the annular groove.

(85) In the above-described pressure gauge, the pressure gauge may include the retaining member having a line shape and configured to prevent the attachment convex portion from coming out of the attachment hole, the outer attachment surface may include the annular groove extending in the circumferential direction about the axis of the attachment convex portion, the adapter may include the first hole (104) extending linearly from the outer surface (80a) of the adapter to the inner attachment surface, and the second hole (106) located on the extension line of the first hole and extending linearly from the inner attachment surface, the second hole may open to the flat surface (100b) of the inner attachment surface, the flat surface extending in the direction inclined with respect to the extending direction of the second hole, and the retaining member may be inserted into the first hole, the annular groove, and the second hole so as to straddle the first hole, the annular groove, and the second hole.

(86) The above embodiment discloses the fluid pressure system (10) including the fluid pressure device, and the pressure gauge attached to the fluid pressure device, wherein the pressure gauge is the above-described pressure gauge.

Claims

1. A pressure gauge comprising: a case configured to accommodate a measurement unit configured to measure a pressure of a fluid; a display unit provided in the case and configured to display the pressure measured by the measurement unit; and an adapter attached to a fluid pressure device, wherein the adapter includes an attachment hole, the case includes an attachment convex portion configured to be fitted into the attachment hole, the display unit includes a display surface oriented in a direction intersecting an axis of the attachment convex portion, and the attachment convex portion and the attachment hole have a fitting structure that allows a relative position between the attachment convex portion and the adapter in a circumferential direction about the axis of the attachment convex portion to be changed, wherein the adapter includes an inner attachment surface configured to surround the attachment hole, the attachment convex portion includes an outer attachment surface having a polygonal shape, and the inner attachment surface restricts rotation of the attachment convex portion about the axis relative to the adapter in a fitted state in which the attachment convex portion is fitted into the attachment hole.

2. The pressure gauge according to claim 1, wherein the inner attachment surface is formed in a polygonal shape.

3. The pressure gauge according to claim 2, wherein the attachment convex portion is 45° rotationally symmetric about the axis of the attachment convex portion.
4. The pressure gauge according to claim 3, wherein the outer attachment surface is formed in an octagonal shape, and the inner attachment surface includes a plurality of flat surfaces facing a plurality of flat surfaces of the outer attachment surface in the fitted state.
5. The pressure gauge according to claim 4, further comprising a retaining member having a line shape and configured to prevent the attachment convex portion from coming out of the attachment hole, wherein the outer attachment surface includes an annular groove extending in the circumferential direction about the axis of the attachment convex portion, the adapter includes: a first hole extending linearly from an outer surface of the adapter to the inner attachment surface; and a second hole located on an extension line of the first hole and extending linearly from the inner attachment surface, the second hole opens to a flat surface of the inner attachment surface, the flat surface extending in a direction inclined with respect to an extending direction of the second hole, and the retaining member is inserted into the first hole, the annular groove, and the second hole so as to straddle the first hole, the annular groove, and the second hole.
6. The pressure gauge according to claim 1, further comprising a retaining member having a line shape and configured to prevent the attachment convex portion from coming out of the attachment hole, wherein the outer attachment surface includes an annular groove extending in the circumferential direction about the axis of the attachment convex portion, the adapter includes an insertion hole communicating with the annular groove in the fitted state, and the retaining member is inserted into the insertion hole and the annular groove so as to straddle the insertion hole and the annular groove.
7. A fluid pressure system comprising: a fluid pressure device; and a pressure gauge attached to the fluid pressure device, wherein the pressure gauge includes: a case configured to accommodate a measurement unit configured to measure a pressure of a fluid; a display unit provided in the case and configured to display the pressure measured by the measurement unit; and an adapter attached to the fluid pressure device, the adapter includes an attachment hole, the case includes an attachment convex portion configured to be fitted into the attachment hole, the display unit includes a display surface oriented in a direction intersecting an axis of the attachment convex portion, and the attachment convex portion and the attachment hole have a fitting structure that allows a relative position between the attachment convex portion and the adapter in a circumferential direction about the axis of the attachment convex portion to be changed, wherein the adapter includes an inner attachment surface configured to surround the attachment hole, the attachment convex portion includes an outer attachment surface having a polygonal shape, and the inner attachment surface restricts rotation of the attachment convex portion about the axis relative to the adapter in a fitted state in which the attachment convex portion is fitted into the attachment hole.
8. The fluid pressure system according to claim 7, wherein the inner attachment surface is formed in a polygonal shape.
9. The fluid pressure system according to claim 8, wherein the attachment convex portion is 45° rotationally symmetric about the axis of the attachment convex portion.
10. The fluid pressure system according to claim 9, wherein the outer attachment surface is formed in an octagonal shape, and the inner attachment surface includes a plurality of flat surfaces facing a plurality of flat surfaces of the outer attachment surface in the fitted state.
11. The fluid pressure system according to claim 10, wherein the pressure gauge includes a retaining member having a line shape and configured to prevent the attachment convex portion from coming out of the attachment hole, the outer attachment surface includes an annular groove extending in the circumferential direction about the axis of the attachment convex portion, the adapter includes: a first hole extending linearly from an outer surface of the adapter to the inner attachment surface; and a second hole located on an extension line of the first hole and extending

linearly from the inner attachment surface, the second hole opens to a flat surface of the inner attachment surface, the flat surface extending in a direction inclined with respect to an extending direction of the second hole, and the retaining member is inserted into the first hole, the annular groove, and the second hole so as to straddle the first hole, the annular groove, and the second hole.

12. The fluid pressure system according to claim 7, wherein the pressure gauge includes a retaining member having a line shape and configured to prevent the attachment convex portion from coming out of the attachment hole, the outer attachment surface includes an annular groove extending in the circumferential direction about the axis of the attachment convex portion, the adapter includes an insertion hole communicating with the annular groove in the fitted state, and the retaining member is inserted into the insertion hole and the annular groove so as to straddle the insertion hole and the annular groove.
